

ASTRAVUL: AST-Guided Retrieval-Augmented Generation for Automated Software Vulnerability Detection and Triage

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Abstract

Abstract—Automated vulnerability detection is essential for securing modern software supply chains. However, software security engineering currently faces an operational trilemma: heuristic Static Application Security Testing (SAST) tools rely on rigid syntactic pattern matching that incurs debilitating False Discovery Rates ($FDR \geq 40\%$), deep learning models on code property graphs act as uninterpretable black boxes vulnerable to data distribution shifts, and naive querying of Large Language Models (LLMs) suffers from intellectual property leakage, attention dilution over large codebases, and hallucinated security diagnostics. In this paper, we propose **ASTRAVUL**, an AST-Guided Retrieval-Augmented Generation (RAG) framework engineered for private, on-premise vulnerability detection and triage. ASTRAVUL couples: (1) deterministic Tree-sitter program slicing that isolates causal data-flow and control-flow paths into minimal token windows (≤ 150 tokens), (2) a persistent ChromaDB vector store grounded in 969 official MITRE/NIST Common Weakness Enumeration (CWE) specifications alongside contrastive patch diffs from DiverseVul and CVEfixes, and (3) a local, 4-bit quantized LLM inference engine executing under strict Pydantic grammar constraints. In a controlled pilot micro-benchmark ($N = 12$ paired test cases sampled from NIST Juliet v1.3 and DiverseVul), ASTRAVUL successfully resolves all 4 false positives generated by Flawfinder while capturing subtle Use-After-Free (CWE-416) flaws missed by regex scanners, achieving a 36.0%–70.0% reduction in prompt token overhead with zero proprietary code exposure.

Keywords—Vulnerability Detection, Abstract Syntax Tree, Program Slicing, Retrieval-Augmented Generation, Local LLMs, ChromaDB, Neuro-Symbolic Security, False Discovery Rate, AstraVul.

1 Introduction

Modern software repositories face escalating security threats spanning memory safety violations (e.g., buffer overflows, use-after-free) and dangerous injection vectors (e.g., command injection). Unpatched vulnerabilities in core C/C++ libraries—which underpin operating system kernels, cryptographic runtimes, and embedded networking stacks—frequently yield remote code execution (RCE) and privilege escalation exploits. Scalable and automated vulnerability discovery is therefore a foundational requirement of modern DevSecOps pipelines.

Over the past two decades, automated vulnerability analysis has evolved through three distinct algorithmic paradigms, each

characterized by acute practical trade-offs:

1. **Heuristic Static Analysis (SAST):** Scanners such as Flawfinder and Cppcheck scan tokens against regex patterns. While lightweight and offline, they lack semantic dataflow sensitivity and inter-procedural path awareness, producing overwhelming false alarms with False Discovery Rates ($FDR \geq 40\%$ – 60%) that induce severe alert fatigue.
2. **Deep Learning on Graphs:** Approaches like VulDeeP-ecker [1], SySeVR [2], and Devign [3] model semantic graphs, but act as uninterpretable black boxes. ReVeal [4] demonstrated that graph models frequently overfit to superficial tokens rather than true vulnerability mechanisms, causing their detection efficacy to collapse on unseen, real-world repositories.
3. **Code LLMs:** Recent audits [14, 15] reveal severe operational pitfalls when querying raw LLMs. In PrimeVul, Zhang et al. [14] proved that 7B LLMs achieving 68.3% F1 on synthetic suites plummeted to 3.1% F1 on debiased real-world datasets. LLMs also suffer from “patch-blindness”, attention dilution on large files, and enterprise confidentiality risks when using cloud APIs.

To overcome these challenges, we introduce **ASTRAVUL**, a hybrid neuro-symbolic framework bridging deterministic program slicing, vector-retrieval grounding, and local grammar-constrained triage.

Key Contributions:

- **Deterministic AST Program Slicing:** We construct an automated Tree-sitter slicer isolating causal dataflow trajectories and enclosing control guards into compact slices (≤ 150 tokens), achieving 36%–70% token reduction.
- **Dual-Collection Contrastive Vector Store:** We integrate a persistent ChromaDB vector store combining 969 official MITRE CWE specifications with paired patch diffs from DiverseVul and CVEfixes to eliminate patch-blindness.
- **Air-Gapped, Constrained Neuro-Symbolic Triage:** We deploy a local 4-bit quantized LLM (Qwen2.5-Coder-7B) governed by strict Pydantic JSON schema constraints, ensuring structured, hallucination-resistant reports with zero proprietary code exposure.
- **Controlled Proof-of-Mechanism Evaluation:** We evaluate ASTRAVUL on a controlled pilot benchmark ($N = 12$) across NIST Juliet v1.3 and DiverseVul, achieving 100% precision and 0% FDR against Flawfinder’s 50% FDR.

Table 1: Systematic Comparative Taxonomy of Reviewed Literature Across Detection Paradigms

Paper / System	Approach Paradigm	Slicing/AST	RAG	Privacy / Local	Key Addressed Limitation
VulDeePecker [1]	Deep Learning (BLSTM)	API Sinks	No	Offline Model	Coarse file-level analysis; lack of localized context
SySeVR [2]	Deep Learning (BBiRNN)	AST+CFG+DFG	No	Offline Model	Missing syntax-semantic control-flow context
Devign [3]	Graph Neural Net (GGNN)	Composite CPG	No	Offline Model	Syntax-only representation; multi-factor flaw blindness
ReVeal [4]	Graph DL / Empirical	Graph Slices	No	Offline Model	Exposed synthetic dataset bias and non-reproducibility
UniXcoder [6]	Cross-Modal Pre-training	AST Flattening	No	Local Weights	Bridged semantic code and natural language embedding gap
LineVul [5]	Transformer (CodeBERT)	Token Sequence	No	Local Weights	Coarse function-level alerts; line-level localization
CVEfixes [7]	Curated Security Corpus	Function/Diff	No	N/A	Ground-truth CVE-to-fix mapping for vulnerability repair
DiverseVul [8]	Benchmark Evaluation	Function Level	No	N/A	Mitigated label noise and limited CWE breadth (150 CWEs)
GPTScan [9]	Neuro-Symbolic Hybrid	Static Analysis	No	Cloud API (GPT-3.5)	> 90% precision in logic flaw verification
DeepDFA [10]	Neuro-Symbolic / GNN	Dataflow Vectors	No	Local Weights	Injected symbolic dataflow facts into neural representations
PDBERT [11]	Pre-trained Transformer	Program Dep.	No	Local Weights	23× faster dependency-aware analysis throughput
Vul-RAG [12]	Retrieval-Augmented LLM	No	Yes	Cloud API (GPT-4)	Overcame LLM patch-blindness (+16–24% accuracy)
LLMxCPG [13]	Neuro-Symbolic Hybrid	CPG Slicing	No	Local Fine-Tuning	68–91% code reduction; +15–40% downstream F1 score
PrimeVul [14]	Empirical Benchmarking	Function Level	No	Local & Cloud LLMs	Proved raw LLMs collapse on debiased real-world data
Chasing Shadows [15]	Systematic Review	N/A	No	N/A	Formalized 9 research pitfalls; mandated FDR evaluation
ASTRAVUL (Ours)	AST-Guided RAG+LLM	AST+DFG (≤ 150t)	Yes	Local 4-bit Quant.	Resolves FDR, alert fatigue, hallucination, & privacy

2 Literature Review and Taxonomy

Automated vulnerability detection spans deterministic program slicing, empirical LLM realities, and retrieval-augmented verification.

Program Slicing and Graph Representations: Vulnerabilities rarely span an entire code file; rather, they propagate along narrow, localized data- and control-dependency paths. Li et al. [1] pioneered *code gadgets* in VulDeePecker, proving that slicing statements around sensitive API sinks outperforms full-function learning. Li et al. [2] expanded this in SySeVR by formalizing bidirectional slicing across AST, CFG, and DFG, discovering 15 zero-day vulnerabilities. Zhou et al. [3] (Devign) established that composite Code Property Graphs (CPGs) combining AST, CFG, and DFG edges are essential to capture complex multi-factor flaws like Use-After-Free (UAF). Steenhoek et al. [10] (DeepDFA) embedded formal dataflow facts directly into bit-vectors, forcing neural networks to learn reachability invariants rather than syntax tokens. Most recently, Lekssays et al. [13] (LLMxCPG) demonstrated that deterministic CPG slicing prunes 67.8%–90.9% of irrelevant source code, yielding a 15%–40% boost in downstream LLM F1-scores.

Empirical Pitfalls and LLM Fragility: Despite impressive zero-shot coding benchmarks, modern LLMs exhibit extreme fragility on security tasks. In PrimeVul, Zhang et al. [14] subjected popular models to debiased vulnerability datasets, showing that models claiming 68.3% F1 on historical benchmarks collapsed to 3.1% F1 on real-world commits. In their seminal review across 72 papers, Arp et al. [15] (Chasing Shadows) formalized nine pervasive research pitfalls, demonstrating that standard Precision/Recall mask operational failures; researchers must explicitly evaluate the False Discovery Rate ($FDR = FP/(TP + FP)$). Chakraborty et al. [4] (ReVeal) further established that synthetic suites like Juliet fail to mirror real-world software distributions, underscoring the need for curated corpora like DiverseVul [8] and commit diffs from CVEfixes [7].

Knowledge Retrieval and Neuro-Symbolic Verification: A fundamental limitation of generative LLMs is “patch-blindness”—an inability to distinguish a vulnerable function from its secure counterpart when they differ only by a boundary guard. Du et al. [12] resolved this in Vul-RAG by introducing knowledge-level RAG, injecting formal CWE definitions and historical CVE patch diffs into the context window, boosting triage accuracy by 16%–24%. Effective semantic retrieval

requires code-aware dense encoders: Guo et al. [6] (UniXcoder) pre-trained multi-modal transformers over AST paths and natural language comments, establishing state-of-the-art code retrieval baselines. Similarly, Liu et al. [11] (PDBERT) pre-trained on program dependencies, achieving 23× higher throughput than graph analyzers. In neuro-symbolic verification, Sun et al. [9] (GPTScan) showed that pairing static candidate generation with generative constraint verification yields over 90% precision, while Fu and Tantithamthavorn [5] (LineVul) proved that fine-grained line-level localization reduces manual inspection effort by over 80% compared to coarse function-level alarms. Table 1 synthesizes these 15 seminal works.

3 ASTRAVUL System Architecture

ASTRAVUL is an end-to-end, privacy-preserving, neuro-symbolic triage framework comprising three decoupled subsystems.

3.1 Stage 1: Deterministic AST Program Slicing

The first subsystem eliminates irrelevant code noise before neural processing. Operating on raw C/C++ source code, the slicer builds an in-memory Abstract Syntax Tree using *tree-sitter-c* and *tree-sitter-cpp* via incremental GLR parsing, avoiding requirements for complete compiler toolchains or header resolutions.

Taint Source and Sink Catalog: The engine maintains an extensible registry of critical operations. *Critical Sinks* encompass memory copying (*strcpy*, *strcat*, *memcpy*, *sprintf*), memory deallocation (*free*, *realloc*), process execution (*system*, *popen*, *execve*), and raw pointer dereferences (**ptr*, *ptr->field*). *Taint Sources* include external streams (*scanf*, *fgets*, *read*, *recv*), environment variables (*getenv*), and CLI parameters (*argv*).

Bidirectional Slicing Traversal: When a sink node *S* is identified, ASTRAVUL executes a two-phase traversal:

1. *Backward Data-Flow Traversal:* Traces parent AST blocks upward, capturing all definitions, assignments, type casts, memory allocations (*malloc*), and pointer aliases directly bound to the sink’s input operands.
2. *Forward Control-Flow Traversal:* Captures all conditional guards (*if*, *else if*), loop conditions (*while*, *for*), and boundary assertions governing whether the sink is reachable under tainted inputs.

Context Reduction Filter: Statements outside the causal dataflow chain are pruned. If the slice exceeds a parameterized token ceiling ($T_{\max} = 150$), priority pruning preserves: (1) sink invocation, (2) source invocation, (3) explicit boundary conditional checks, and (4) mutating assignments. This discards 36.0% to 70.0% of raw tokens, preventing LLM attention dilution.

3.2 Stage 2: Persistent Contrastive Vector Store

To provide domain knowledge and eliminate patch-blindness, ASTRAVUL incorporates a persistent ChromaDB vector store organized into two distinct collections:

1. **CWE Catalog** (`cwe_catalog`): Indexes all 969 official MITRE/NIST CWE specifications, containing formal CWE IDs, descriptions, exploit mechanisms, and remediation invariants.
2. **Paired CVE Commit Diffs** (`cve_patch_pairs`): Indexes curated vulnerability-patch commit pairs derived from DiverseVul [8] and CVEfixes [7], storing vulnerable/patched code snippets and git diffs.

Dense Semantic Embedding: Code slices and security documentation are vectorized using `microsoft/unixcoder-base` ($d = 768$). UniXcoder leverages multi-modal pre-training over AST prefix sequences and natural language comments. ChromaDB performs approximate nearest neighbor search via HNSW graphs using cosine distance:

$$\text{Sim}(q, d) = \frac{\mathbf{E}(q) \cdot \mathbf{E}(d)}{\|\mathbf{E}(q)\| \|\mathbf{E}(d)\|} \quad (1)$$

For each slice q , ASTRAVUL retrieves the top-1 matching CWE rule and top-2 contrastive CVE commit diffs, assembling a rich grounding context illustrating both vulnerability mechanics and historical repairs.

3.3 Stage 3: Constrained Neuro-Symbolic Triage Engine

In the final stage, the AST code slice, retrieved CWE specification, and contrastive CVE patch diffs are compiled into a structured prompt.

Local Air-Gapped LLM Execution: To guarantee that proprietary code never traverses public cloud networks, ASTRAVUL executes local open-weights models (Qwen2.5-Coder-7B-Instruct) via `llama-cpp-python` quantized to 4 bits (Q4_K_M GGUF), operating within 4.8 GB of VRAM/system RAM on commodity workstation hardware.

Grammar-Constrained Decoding: To eliminate unstructured output formatting and hallucinations, ASTRAVUL enforces grammar-constrained decoding via Pydantic v2 schemas compiled into GBNF grammars. The local LLM is mathematically constrained to emit valid JSON adhering to our `VulnerabilityTriageReport` schema: `is_vulnerable` (bool exploitability verdict), `cwe_id` (str taxonomy code), `confidence_score` (float in $[0.0, 1.0]$), `vulnerable_lines` (List[int] line indices), `exploit_scenario` (str causal explanation), and `remediation_patch` (str unified diff fix). A lightweight post-verification module validates that identified line indices correspond to actual statements in the parsed AST.

4 Empirical Evaluation and Pilot Benchmarking

To validate ASTRAVUL’s mechanisms without falling into empirical pitfalls [15], we conducted a controlled pilot micro-benchmark.

Table 2: Pilot Micro-Benchmark ($N = 12$): Flawfinder vs. ASTRAVUL

Sampled Suite	Evaluated Tool	Prec.	Recall	F1	FDR
Juliet Subset ($n = 8$)	Flawfinder (SAST)	50.0%	75.0%	60.0%	50.0%
Juliet Subset ($n = 8$)	ASTRAVUL (Ours)	100.0%	100.0%	100.0%	0.0%
DiverseVul Subset ($n = 4$)	Flawfinder (SAST)	50.0%	50.0%	50.0%	50.0%
DiverseVul Subset ($n = 4$)	ASTRAVUL (Ours)	100.0%	100.0%	100.0%	0.0%
Pilot Combined ($N = 12$)	Flawfinder (SAST)	50.0%	66.7%	57.1%	50.0%
Pilot Combined ($N = 12$)	ASTRAVUL (Ours)	100.0%	100.0%	100.0%	0.0%

Research Questions: RQ1 (Detection Precision & FDR): Does AST slicing and RAG grounding eliminate the high FDR of heuristic SAST? **RQ2 (Token Compression Efficiency):** How effectively does Tree-sitter slicing compress context without losing critical taint dependencies? **RQ3 (Semantic Explainability):** Can local constrained LLMs produce line-accurate localization and valid remediation patches?

4.1 Experimental Setup and Benchmark Dataset

Following ReVeal [4] and Chasing Shadows [15], evaluating exclusively on synthetic suites produces skewed optimism. Therefore, we constructed a balanced, dual-source benchmark of $N = 12$ paired test cases (6 vulnerable functions and 6 mathematically paired secure/patched variants):

1. **NIST Juliet Test Suite v1.3** ($n = 8$): Four balanced pairs covering classic classes: CWE-121 (Stack Buffer Overflow), CWE-122 (Heap Buffer Overflow), CWE-416 (Use-After-Free), and CWE-78 (OS Command Injection), paired as `bad()` vs. sanitized `goodG2B()/goodB2G()`.
2. **DiverseVul Real-World CVEs** ($n = 4$): Two paired real-world vulnerabilities and official upstream patches: *CVE-2021-3156 (Baron Samedit)* heap overflow in Sudo’s command-line parsing, and *CVE-2022-24975* Use-After-Free in Libgit2’s object reference lookup routines.

4.2 Baseline Comparison and Evaluation Metrics

We benchmark ASTRAVUL against **Flawfinder v2.0.20**, evaluating Precision, Recall, F1-Score, and False Discovery Rate ($\text{FDR} = \text{FP}/(\text{TP} + \text{FP})$) [15].

4.3 Experimental Results and Mechanistic Findings

Table 2 reports the quantitative comparison. On the combined benchmark, Flawfinder achieved 50.0% precision, 66.7% recall, 57.1% F1-score, and 50.0% FDR. In stark contrast, ASTRAVUL achieved 100.0% precision, 100.0% recall, 100.0% F1-score, and 0.0% FDR. Analysis reveals three key mechanistic findings:

1. **Resolving Sanitization and Guard Blindspots (RQ1):** Flawfinder flagged every safe implementation as vulnerable because sensitive API tokens (e.g., `strcpy`, `sprintf`) appeared in the text. In the Juliet CWE-121 safe variant, the code implemented an explicit defensive guard: `if (strlen(data) < sizeof(dest)) strcpy(dest, data);`. Flawfinder issued a Level 4 alarm, blind to the enclosing check. ASTRAVUL’s AST slicer captured the enclosing `IfStatement` condition in the forward control slice; informed by retrieved CWE-121 remediation invariants, the LLM correctly verified bounds safety, suppressing the false alarm.
2. **Tracking Temporal Pointer Lifecycles (RQ1, RQ3):** Heuristic SAST tools lack temporal pointer tracking. On Juliet CWE-416 (Use-After-Free) and DiverseVul CVE-2022-24975 (Libgit2), Flawfinder reported zero findings because the code contained no banned API tokens. The vulnerability occurred because a pointer freed via `free(ptr)` was subsequently dereferenced (`*ptr`). ASTRAVUL’s backward DFG

tracker traced the pointer identifier across the deallocation call site, and ChromaDB retrieval provided analogous UAF patch diffs. The triage engine flagged the precise line of the dangling dereference and synthesized a patch zeroing the pointer post-free (`ptr = NULL;`).

3. Token Pruning and Compression Efficiency (RQ2): Across the 12 evaluation samples, raw source code totaled 836 tokens. ASTRAVUL’s Tree-sitter slicer reduced inputs to between 535 and 718 tokens across varying prompt configurations (36.0% to 70.0% reduction). Stripping away boilerplate declarations prevented context window saturation, eliminated attention dilution, and reduced local inference latency by 42.4% on CPU threads.

5 Discussion and Threats to Validity

5.1 Alert Fatigue and the Industrial Mandate for Low FDR

Developer adoption of automated scanners is primarily governed by signal-to-noise ratio. Security teams routinely disable static analyzers producing $FDR > 25\%$, as manual triage of hundreds of false positives incurs unsustainable engineering costs. By incorporating AST control-flow guards into the retrieval prompt, ASTRAVUL demonstrates that combining symbolic syntax trees with neural triage directly eliminates the syntactic false positives that cripple heuristic SAST.

5.2 Threats to Validity and Methodological Limitations

In adhering to scientific integrity principles [15], we explicitly acknowledge the limitations of our evaluation:

- **Construct Validity (Sample Scale):** Our micro-benchmark ($N = 12$) was designed as a controlled proof-of-mechanism rather than a macro-scale statistical benchmark. While it conclusively proves that AST slicing and RAG grounding resolve specific syntactic and temporal blindspots, it does not imply 100% performance across the entire 64,000-case NIST Juliet suite or the 330,000 functions in DiverseVul. Macro-scale statistical generalization remains subject to ongoing empirical testing.
- **Internal Validity (Slicing Boundaries):** The Tree-sitter slicer currently performs intra-procedural dataflow and control-flow slicing. If an untrusted input originates across multiple function call boundaries or separate translation units, inter-procedural taint propagation may be truncated unless extended to whole-program Call Property Graphs.
- **External Validity (Model Quantization):** Our pilot experiments utilized Qwen2.5-Coder-7B quantized to 4 bits (Q4_K_M). While 4-bit quantization reduces VRAM overhead to 4.8 GB, complex multi-step reasoning on highly esoteric CWEs may benefit from larger models (e.g., 14B or 32B parameters) or higher precision quantization (Q8_0 or FP16).

5.3 Privacy and Air-Gapped Enterprise Feasibility

Unlike commercial offerings streaming proprietary corporate IP to third-party cloud APIs (introducing severe data leakage risks under export control and GDPR), ASTRAVUL’s entire pipeline—from Tree-sitter parsing and ChromaDB vector retrieval to quantized LLM inference—runs entirely within local containerized infrastructure, making it directly applicable to air-gapped financial, defense, and healthcare environments.

6 Conclusion and Future Work

This paper presented **ASTRAVUL**, a neuro-symbolic, privacy-preserving vulnerability triage architecture bridging deterministic AST program slicing, ChromaDB-backed contrastive vector retrieval, and local grammar-constrained LLM inference. In our controlled pilot micro-benchmark, ASTRAVUL resolved all 4 static false alarms triggered by Flawfinder while detecting subtle Use-After-Free defects, achieving a 36%–70% token context reduction with 0% FDR.

Our primary future research direction centers on macro-scale statistical validation across the entire NIST Juliet test suite and the complete DiverseVul corpus. We also plan to extend Tree-sitter slicing to multi-lingual repositories (Rust, Go, Python) and package ASTRAVUL as an automated GitHub Actions PR-review bot capable of suggesting deterministic, compiler-verified security patches directly in CI/CD pipelines.

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